

Hedra Emad Fawzy

Full-Stack Developer | TypeScript, NestJS, React & Next.js

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PROFESSIONAL SUMMARY

Full-Stack Developer specializing in TypeScript across the Node.js and React ecosystems. Led a 5-developer team as Team Leader on EduGenie, an AI-powered e-learning platform built and deployed in six weeks, where I architected a NestJS + MongoDB backend of 188 REST endpoints, built a Retrieval-Augmented Generation (RAG) pipeline, and integrated Stripe Connect payments end to end. Trained through the ITI Intensive Code Camp (618 hours, Full-Stack Web & Generative AI using MERN) and hold a B.Sc. in Computer Science. Looking for a Full-Stack or Backend Developer role on a product team.

TECHNICAL SKILLS

Proficient: TypeScript, JavaScript (ES6+), Node.js, NestJS, Express.js, React.js, Next.js, MongoDB & Mongoose, REST API Design, Jest Unit Testing, Git & GitHub

Working Knowledge: GraphQL & Apollo Client, SQL Server, Redux Toolkit, TanStack Query, Tailwind CSS, Socket.IO & WebSockets, Stripe API, Cloud Deployment (Vercel, Railway)

AI Engineering: Retrieval-Augmented Generation (RAG), Embeddings & Semantic Search, OpenAI API, Google Gemini API, Prompt Engineering

Practices: Role-Based Access Control, JWT & OAuth 2.0, Secure Coding, API Documentation (Swagger), Code Review, Agile / Scrum, ESLint & Prettier

PROJECTS

EduGenie — AI-Powered E-Learning Platform | Team Leader & Full-Stack Developer

ITI Intensive Code Camp | Team of 5 | May – Jul 2026 (6-week build)

Live: [edugenie-student-web.vercel.app](#) · [edugenie-dashboard.vercel.app](#) · [edugenie-api.vercel.app](#)

Code: [github.com/hedra-emad/edugenie-api](#) · [/edugenie-student-web](#) · [/edugenie-dashboard](#)

- Led a 5-developer team as Team Leader, owning architecture decisions, task distribution and code review, and shipping all three applications to production.
- Architected the backend in NestJS, TypeScript and MongoDB: 188 REST endpoints across 32 feature modules (auth, courses, enrollments, quizzes, payments, certificates, notifications, reporting), documented with Swagger and covered by a green Jest suite of 33 test files / 140 tests across services, controllers and the RAG layer.
- Built a Retrieval-Augmented Generation (RAG) pipeline — document chunking, embeddings, indexing and semantic retrieval — powering an AI study coach, auto-generated learning roadmaps and practice/remediation flows.
- Integrated the OpenAI and Google Gemini APIs behind a swappable provider interface, so either model can be used for lecture transcription without changing application code.
- Implemented end-to-end payments with Stripe Connect (Express accounts, destination charges, signature-verified webhooks), handling student checkout and automated instructor payouts.
- Secured the API with JWT and Google OAuth 2.0, role-based access control across four roles (student, instructor, admin, super-admin), Helmet, request rate limiting and DTO-level validation.
- Delivered real-time notifications and live AI chat over Socket.IO and Pusher WebSocket gateways.
- Built the student-facing app in Next.js and React — 28 routes and 160+ components using TanStack Query, React Hook Form with Zod, and Framer Motion.

Optern — Virtual Job Preparation Platform | Front-End Developer

University Graduation Project | Team of 4 | Nov 2024 – Jun 2025 | [github.com/hedra-emad/Optern-Frontend](#)

- Delivered 48 merged pull requests as one of four developers on a Next.js and TypeScript platform offering AI-driven mock interviews, an AI CV builder, and virtual collaboration rooms.
- Built the blog module, user profile, notifications, and the complete authentication flow (login, registration, email verification, password reset).
- Developed the collaboration-room features — room creation and join requests, workspace and sprint boards — plus the shared component library reused across the application.
- Consumed GraphQL APIs via Apollo Client with Redux Toolkit state management, working within a layered domain / application / infrastructure / presentation architecture and a pull-request review workflow.

EMS — Examination Management System — Front-End Developer | South Valley University | Feb – Jun 2024

- Built responsive, cross-browser exam-management interfaces for a university internship project.

TRAINING

Intensive Code Camp – Full-Stack Web & Generative AI Development using MERN

Information Technology Institute (ITI) | 618 Hours | 2025 – 2026

- Project-based training in TypeScript, React.js, Next.js, Node.js, NestJS, GraphQL, MongoDB, unit testing, design patterns, secure coding, and Generative AI development.

Full-Stack Web Development Diploma

Route Academy | 9 Months | Aug 2022 – May 2023

- Intensive diploma covering front-end and back-end web development fundamentals, databases, and hands-on project work.

EDUCATION

B.Sc. in Computer Science and Information Technology

Qena University (South Valley University), Faculty of Computers and Information | Oct 2021 – Jul 2025

CERTIFICATIONS & ACTIVITIES

- Learning How to Learn — Deep Teaching Solutions (Coursera), Feb 2024 | coursera.org/verify/QWBTERUS6NCQ
- Freelancing Fundamentals — Mahara Tech (ITI), Feb 2026: Freelancing Basics plus platform-specific modules for Upwork, Mostaqi and Khamsat.
- Public Relations Member, ACPC Community — Qena University | 2021 – 2023: organized and promoted competitive-programming events and workshops for the student developer community.

ADDITIONAL EXPERIENCE

Accountant, part-time (2026 – Present) — project financing, installment collection, and data entry into the company system.
Freelance Videographer & Video Editor (2022 – 2025) — produced and edited promotional video content using Adobe Premiere Pro and Lightroom.

LANGUAGES

Arabic: Native | English: B2 (Upper-Intermediate)